NATE POWELL

SOFTWARE ENGINEERING / AI SYSTEMS / DATA TOOLING

Waterloo, ON | n2powell@uwaterloo.ca | github.com/Tophacks | quadmates.ca

PROFILE

Product-focused software developer and founder who ships end-to-end web applications, builds Python simulation and data tools, and evaluates reinforcement-learning systems. Incoming Infosys InStep intern in AI Agent Engineering (January 2027).

EXPERIENCE

Quadmates | Founder & Full-Stack Engineer

2026 - Present

- Built and deployed a student networking platform using React, Vite, Node.js/Express, PostgreSQL and Supabase, including course matching, study groups, real-time messaging and time-based study tracking.
- Developed REST APIs, JWT-protected account flows, database migrations and privacy/safety controls; integrated Supabase Realtime and user-facing reporting workflows.
- Iterated on product reliability and responsive UX, and prepared an iOS prototype using Capacitor and Vite.

Infosys InStep | Incoming Computer Engineering Intern

Jan - Apr 2027

- Selected for the CTO office in Bengaluru, India; anticipated focus on AI agent engineering and enterprise workflow automation. Internship has not started.

UW Orbital | Business Communications & Engineering Support

2026 - Present

- Negotiated continued corporate manufacturing support by aligning CubeSat engineering drawings and sponsor production schedules while preserving the long-term relationship.
- Organized engineering expense data in Excel and prototyped Python workflows for reporting and automation.

SELECTED ENGINEERING PROJECTS

TransitOps | Python, SUMO/TraCI, FastAPI, Next.js

Personal project

- Developed a deterministic transit-operations simulator comparing dispatch strategies, passenger queues and disruption scenarios; integrated SUMO vehicle/signal telemetry through FastAPI WebSockets to a MapLibre dashboard.
- Added Docker-based simulator checks and GitHub Actions build/smoke-test validation.

NeuroSim 2.0 | Python, PPO, Stable-Baselines3

Personal project

- Built a 2D rocket-landing control environment for reinforcement-learning experiments, supporting continuous actions, flight recording and policy evaluation on held-out scenarios.

TECHNICAL SKILLS

Languages	Python, JavaScript, TypeScript, SQL, Java
Applications / Data	React, Vite, Next.js, Node.js, Express, FastAPI, PostgreSQL, Supabase, REST APIs, WebSockets
AI / Infrastructure	PyTorch, PPO, Stable-Baselines3, OpenCV, Docker, Git, GitHub Actions, Vercel, Capacitor

EDUCATION

University of Waterloo

2025 - Present

Honours Mathematical Physics (Co-op); University of Waterloo